

Mustafa Abbas Abokhadher

2240002889

7MA2

Thresholding

```
import cv2

# Load the image in grayscale
img = cv2.imread('Parrot.jpg', cv2.IMREAD_GRAYSCALE)

# Set the threshold values
threshold_values = [0, 50, 100, 150, 200]

cv2.imshow('Original Image', img)

for threshold_value in threshold_values:
    ret, thresh = cv2.threshold(img, threshold_value, 255,
                                cv2.THRESH_BINARY)
    cv2.imshow(f'Thresholded Image ({threshold_value})', thresh)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Histogram Processing

```
import cv2
import matplotlib
import matplotlib.pyplot as plt
import numpy as np
from skimage import data, img_as_float
from skimage import exposure

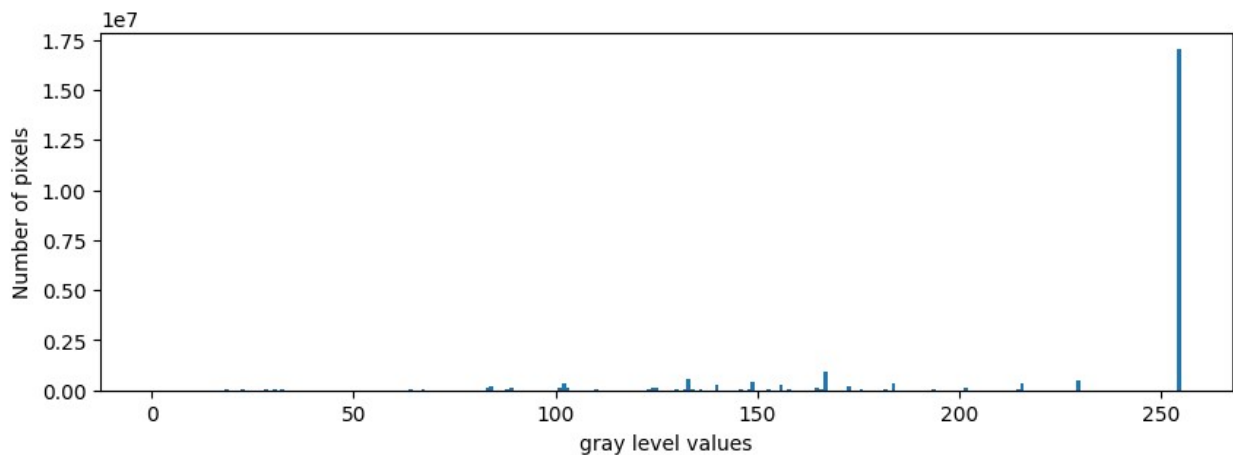
img = cv2.imread('Parrot.jpg', cv2.IMREAD_GRAYSCALE)
p2, p98 = np.percentile(img, (2, 98))
img_rescale = exposure.rescale_intensity(img, in_range=(p2, p98))

fig = plt.figure(figsize=(10, 7))
fig.add_subplot(2, 1, 1)
```

```
plt.imshow(img, cmap = 'gray')
plt.axis('off')
plt.title('Original image')
fig.add_subplot(2, 1, 2)
plt.hist(img.flat, bins = 256, range=(0, 255))
plt.xlabel('gray level values')
plt.ylabel('Number of pixels')
plt.show()

fig = plt.figure(figsize=(10, 7))
fig.add_subplot(2, 1, 1)
plt.imshow(img_rescale, cmap = 'gray')
plt.axis('off')
plt.title('Contrast stretch')
fig.add_subplot(2, 1, 2)
plt.hist(img_rescale.flat, bins = 256, range=(0, 255))
plt.xlabel('gray level values')
plt.ylabel('Number of pixels')
plt.show()
```

Original image



```

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KeyboardInterrupt                                Traceback (most recent call
last)
Cell In[10], line 20
    18 plt.title('Contrast stretch')
    19 fig.add_subplot(2, 1, 2)
--> 20 plt.hist(img_rescale.flat, bins = 256, range=(0, 255))
    21 plt.xlabel('gray level values')
    22 plt.ylabel('Number of pixels')

File c:\Users\musta\AppData\Local\Programs\Python\Python314\Lib\site-
packages\matplotlib\_api\deprecation.py:453, in
make_keyword_only.<locals>.wrapper(*args, **kwargs)
    447 if len(args) > name_idx:
    448     warn_deprecated(
    449         since, message="Passing the %(name)s %(obj_type)s "
    450         "positionally is deprecated since Matplotlib %(
(since)s; the "
    451         "parameter will become keyword-only in %(removal)s.",
    452         name=name, obj_type=f"parameter of {func.__name__}()")
--> 453 return func(*args, **kwargs)

File c:\Users\musta\AppData\Local\Programs\Python\Python314\Lib\site-
packages\matplotlib\pyplot.py:3478, in hist(x, bins, range, density,
weights, cumulative, bottom, histtype, align, orientation, rwidth,
log, color, label, stacked, data, **kwargs)
    3453 @_copy_docstring_and_deprecators(Axes.hist)
    3454 def hist(
    3455     x: ArrayLike | Sequence[ArrayLike],
    (...) 3476     BarContainer | Polygon | list[BarContainer |
Polygon],
    3477 ]:
-> 3478     return gca().hist(
    3479         x,
    3480         bins=bins,
    3481         range=range,
    3482         density=density,
    3483         weights=weights,
    3484         cumulative=cumulative,
    3485         bottom=bottom,
    3486         histtype=histtype,
    3487         align=align,
    3488         orientation=orientation,
    3489         rwidth=rwidth,
    3490         log=log,
    3491         color=color,
    3492         label=label,
    3493         stacked=stacked,
    3494         **({"data": data} if data is not None else {}),

```

```
3495         **kwargs,
3496     )
```

```
File c:\Users\musta\AppData\Local\Programs\Python\Python314\Lib\site-
packages\matplotlib\_api\deprecation.py:453, in
make_keyword_only.<locals>.wrapper(*args, **kwargs)
    447 if len(args) > name_idx:
    448     warn_deprecated(
    449         since, message="Passing the %(name)s %(obj_type)s "
    450         "positionally is deprecated since Matplotlib %
(since)s; the "
    451         "parameter will become keyword-only in %(removal)s.",
    452         name=name, obj_type=f"parameter of {func.__name__}()")
--> 453 return func(*args, **kwargs)
```

```
File c:\Users\musta\AppData\Local\Programs\Python\Python314\Lib\site-
packages\matplotlib\__init__.py:1524, in
_preprocess_data.<locals>.inner(ax, data, *args, **kwargs)
    1521 @functools.wraps(func)
    1522 def inner(ax, *args, data=None, **kwargs):
    1523     if data is None:
-> 1524         return func(
    1525             ax,
    1526             *map(cbook.sanitize_sequence, args),
    1527             **{k: cbook.sanitize_sequence(v) for k, v in
kwargs.items()})
    1529     bound = new_sig.bind(ax, *args, **kwargs)
    1530     auto_label = (bound.arguments.get(label_namer)
    1531                  or bound.kwargs.get(label_namer))
```

```
File c:\Users\musta\AppData\Local\Programs\Python\Python314\Lib\site-
packages\matplotlib\axes\_axes.py:7045, in Axes.hist(self, x, bins,
range, density, weights, cumulative, bottom, histtype, align,
orientation, rwidth, log, color, label, stacked, **kwargs)
    7042     stacked = True
    7044 # Massage 'x' for processing.
-> 7045 x = cbook._reshape_2D(x, 'x')
    7046 nx = len(x) # number of datasets
    7048 # Process unit information. _process_unit_info sets the unit
and
    7049 # converts the first dataset; then we convert each following
dataset
    7050 # one at a time.
```

```
File c:\Users\musta\AppData\Local\Programs\Python\Python314\Lib\site-
packages\matplotlib\cbook.py:1423, in _reshape_2D(X, name)
    1421     if nd > 1:
    1422         raise ValueError(f'{name} must have 2 or fewer
dimensions')
-> 1423     result.append(xi.reshape(-1))
```

```
1425 if is_1d:
1426     # 1D array of scalars: directly return it.
1427     return [np.reshape(result, -1)]
```

KeyboardInterrupt:

Task#1

```
import matplotlib.pyplot as plt
import numpy as np
from skimage import data, exposure

img = data.moon()

p3, p80 = np.percentile(img, (3, 80))

img_rescale = exposure.rescale_intensity(img, in_range=(p3, p80))

fig = plt.figure(figsize=(10, 7))

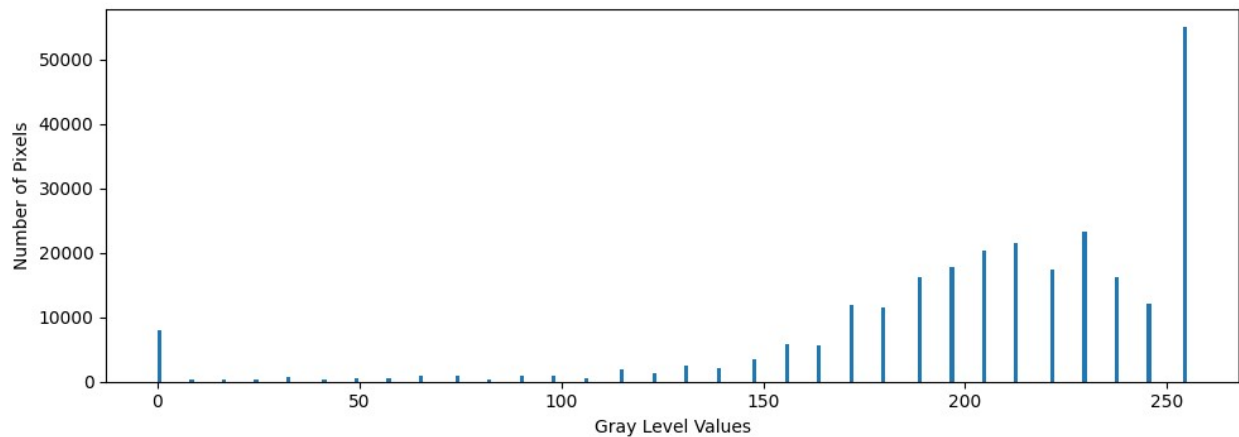
fig.add_subplot(2, 1, 1)
plt.imshow(img_rescale, cmap='gray')
plt.axis('off')
plt.title('Contrast Stretch (3rd & 80th Percentiles)')

fig.add_subplot(2, 1, 2)
plt.hist(img_rescale.flat, bins=256, range=(0, 255))
plt.xlabel('Gray Level Values')
plt.ylabel('Number of Pixels')

plt.tight_layout()
plt.show()

c:\Users\musta\AppData\Local\Programs\Python\Python314\Lib\site-
packages\tqdm\auto.py:21: TqdmWarning: IProgress not found. Please
update jupyter and ipywidgets. See
https://ipywidgets.readthedocs.io/en/stable/user\_install.html
    from .autonotebook import tqdm as notebook_tqdm
```

Contrast Stretch (3rd & 80th Percentiles)



Task#2

```
img = data.moon()

img_eq = exposure.equalize_hist(img)

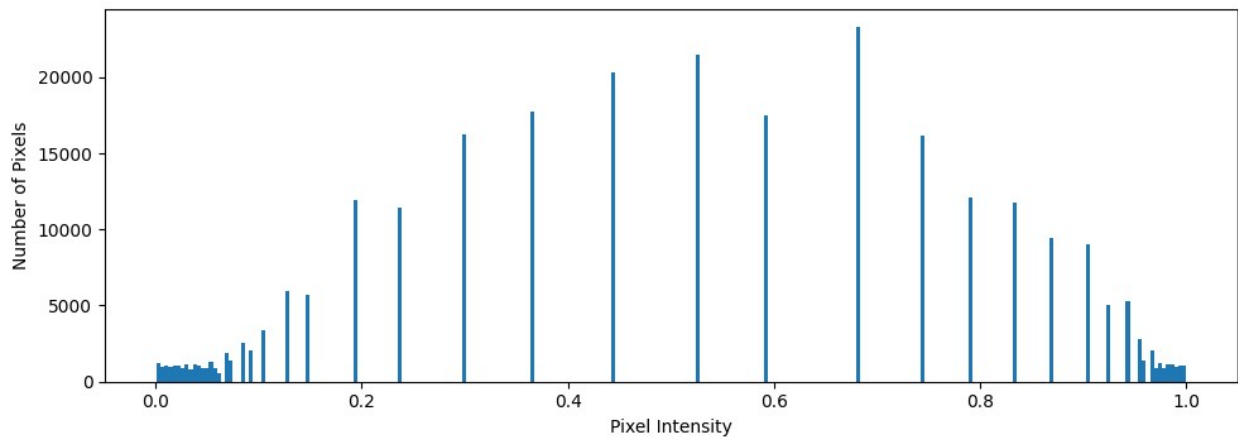
fig = plt.figure(figsize=(10, 7))

fig.add_subplot(2, 1, 1)
plt.imshow(img_eq, cmap='gray')
plt.axis('off')
plt.title('Equalized Image (Flattened Histogram)')

fig.add_subplot(2, 1, 2)
plt.hist(img_eq.flat, bins=256)
plt.xlabel('Pixel Intensity')
plt.ylabel('Number of Pixels')

plt.tight_layout()
plt.show()
```

Equalized Image (Flattened Histogram)



Task#3

```
from skimage.exposure import match_histograms

reference = data.rocket()
image = data.chelsea()

matched = match_histograms(image, reference, channel_axis=-1)

fig, (ax1, ax2, ax3) = plt.subplots(nrows=1, ncols=3, figsize=(15, 5))

ax1.imshow(image)
ax1.set_title('Source Image (Chelsea)')
ax1.axis('off')

ax2.imshow(reference)
ax2.set_title('Reference Image (Rocket)')
ax2.axis('off')

ax3.imshow(matched)
ax3.set_title('Matched Image')
ax3.axis('off')
```

```
plt.tight_layout()  
plt.show()
```

Source Image (Chelsea)



Reference Image (Rocket)



Matched Image

